

An Analysis of Clinical Variables and Gene Expression Data across Biopsies from Patients with Lung Adenocarcinomata

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Introduction

Loading and Cleaning of Annotation and Expression Data

We want to clean up our annotation file. To do this we must first take a look at the file.

We can see here, that there are indeed values missing in a lot of the columns. Looking at the documentation for the data source (`{{SOURCE}}`), we can also see, that the first three segments = or the first 12 digits = of the barcode encode the patient ID. We thus replace `paper_patient` with the barcode trimmed to position 12.

Since we only want the barcode and paper columns, we discard anything not matching to that selection. This leaves us with 20 paper columns.

As this also makes the `'paper_'` prefix superfluous we strip it from the paper columns. Furthermore, we noticed textual NAs, which were entered into the data set as `'[unknown]'`, `'[not available]'` and `'[Not Available]'`. We thus strip these and replace them by real NAs. We perform the same operation on the expression data, whilst we're at it. The `paper_Tumor.stage` column encodes the Ajcc pathological stage. Thus, we replaced it with the ajcc staging.

When looking at the counts of NAs, the expression data are complete, whilst the annotation data have 5623 NAs in total distributed relatively evenly across all columns, with patient and Tumor.stage being the exceptions.

From the head-call above, we know that all of the text columns are characters, even when a factor or numeric type would be more appropriate. We thus go through the cleaned annotation table and factorise, where appropriate. Furthermore, where factorisation is inappropriate, we try to coerce each column to numeric type (accepting both comma and period as decimal delimiters) and discard all failed coercions. Lastly, we double check, that we see the intuitively appropriate classes for each column.

We know from the documentation of the data (`{{SOURCE}}`), that the leading numbers of the fourth segment tell us whether the sample is a tumour or not. We use this to encode a new column entitled `'is.tumour'` storing that information. Since this segment only has the values of `'01'` and `'11'`, and since `'11'` codes for healthy tissue and `'01'` for tumour tissue, a boolean suffices.

With our annotation data (and incidentally also expression data) cleaned up and usable, we move on to the next step.

Data Exploration

Clinical Annotation

Explore pathologic stage data

Explore patient Sex data

There are 110 female and 73 male patients

Explore Smoking status data

Examine Transversion.High.Low and Fusions